Project Outline

RUNTIME TERROR

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1. Introduction

1. Purpose of Plan Document

    -The purpose of this plan document is to break up our advising projects so

it's easier for the team to understand what is going on

2. Project Scope and Objectives

2.1. Statement of Scope- The Student academic progress is a secure web-based application designed to support academic advising and student success. The system will allow students to track their academic progress, monitor degree completion, and receive alerts regarding academic standing. Students will be able to review student records, conduct degree audits, identify at-risk students, and provide academic guidance. This system will focus strictly on academic progress tracking and advising functions. The system is intended for use within a single academic institution and will store and manage student academic records securely while providing role-based access control.

2.2. Major Functions- Role-based access control, Store and display academic history, Track completed and in-progress courses, Calculate and display GPA, Track total earned credits, Define degree requirements by major, Perform automated degree audits, Display completion percentage, identify missing or incomplete requirement, Recommend future courses, Flag students below GPA thresholds, Identify insufficient credit progress, Generate alerts for students, Display warning indicators on dashboards

3. Overview of document- This document outlines the complete project plan for the Student Academic Progress system. It provides a structured explanation of the system's purpose, scope, organization, schedule, risk management strategy, and evaluation criteria.

The document begins with an introduction describing the purpose of the project and defining the system's scope and major functions. It then explains the project organization, including the development process model (Agile methodology), team structure, communication methods, and reporting procedures.

Following that, the document details project resources, including personnel, hardware, and software requirements. The risk management section identifies potential technical, managerial, and user-related risks, along with strategies to reduce or prevent those risks.

The schedule section outlines task allocation, timeline planning, and the task network diagram used to organize workflow. Tracking and control mechanics describe how progress, quality, and scope will be monitored throughout development. An objective grading sheet explains how the project's success will be evaluated.

The document concludes with a glossary of key terms and a list of references used in planning and development.

2. Project Organization

This part is a summary of how we will accomplish this project using agile methodology, as well

as our structure and communication.

2.1.Process Model- We are using the Agile method since it allows us to make prototypes and go back and change things without it being detrimental to our project

2.2.Team Structure- Front end that's in charge of the visual aspect of our website, Backend in charge of the code that allows everything to run smoothly in order for the front end to work , Project Leader that focuses on project scope, timeline, budget, and team coordination , Data Base that makes sure we have all required information in order to have an accurate program(in our case a database of classes) , QA is in charge of our quality assurance that debugs and makes sure our program runs effectively(Calculating GPA, Display of classes, at risk students)

2.3.Communication- IMessage Group chats, in person meetings outside of class (at library), and Email

2.4.Reporting- Once a week in class or out of class

3. Project Resources

This states the people needed to accomplish the project and who will be using it to get this project running, as well as the hardware and software used.

People- **Project Leader** – Oversees scope, timeline, coordination, and ensures deliverables are met.

Frontend Developer(s) – Designs and implements the user interface and dashboard views.

Backend Developer(s) – Develops business logic, GPA calculations, alerts, and system integration.

Database Administrator/Developer – Designs and manages the academic records database (students, courses, majors, requirements).

QA Tester – Tests system functionality (GPA calculation accuracy, alerts, degree audit results, dashboard displays).

Students (End Users) – Use the system to track academic progress and provide usability feedback.

3.1.Hardware and Software- hardware would include laptops and a device to present the project. Software would include GitHub, Visual Studio and python

4. Risk Management

This states every possible risk and how we can manage these issues. 4.1.Likely Risks- incorrect gpa calculations, degree audit errors, integration difficulties, missed deadlines, miscommunication among team, incorrect data, incorrect alerts for at risk students

4.2.Risks Management Plan- conduct code reviews, data validation, provide

simplicity for the student, hold meetings, and split up work evenly.

5. Schedule

This schedule will be on Asana.

5.1. Task List and Resource allocation

- Requirements gathering to define features and scope.
- System design to create wireframes
- Database development to build tables for students, classes, and majors. •

Backend development for gpa, degree audit, and alerts.
• Frontend development, dashboard, and student view of classes. •
QA- Test GPA, alerts, and credit tracking
• Debugging and revisions to fix issues.
• Final review

5.2. Time Line
5.3. Task Network Diagram
5.4. Objective Grading Sheet

6. Tracking and Control Mechanics

Tracking and Control Mechanics describe how the team will monitor progress, maintain quality, and ensure the project stays within scope, time, and performance expectations.

1. Progress Tracking

- Weekly Status Meetings – The team will meet once per week to review completed tasks, upcoming tasks, and any blockers.
- Sprint Reviews (Agile Method) – At the end of each sprint, the team will review completed features (GPA calculator, degree audit, alerts, etc.).
- Task Board (Kanban or To-Do List) – Tasks will be organized into categories such as:
    - To Do
    - In Progress
    - Testing
    - Completed
- Milestone Checkpoints – Major deliverables (Database Setup, Backend Complete, Frontend Complete, Testing Complete) will be reviewed against the timeline.

2. Quality Control

- Code Reviews – Team members will review each other's code to catch errors early.
- QA Testing – The QA member will test:
    - GPA calculation accuracy
    - Credit totals
    - Degree audit results
    - Alert system functionality
- User Testing – Test students/advisors will check usability and clarity of dashboards.
- Bug Tracking Log – Identified issues will be documented and assigned for correction.

3. Scope Control

- All new feature requests must be reviewed by the Project Leader. • Changes must align with the original purpose: academic progress tracking and advising.
- Any scope changes must be documented before implementation to prevent scope creep.

4. Schedule Control

- Compare weekly progress to the original timeline. •
If delays occur:
    - Reassign tasks if necessary
    - Adjust sprint priorities
    - Increase collaboration sessions

5. Performance Metrics

The project will be evaluated using measurable outcomes:

- Accurate GPA calculation (100% formula correctness) • Correct credit tracking
- Accurate degree completion percentage
- Proper alert generation for at-risk students
- Minimal system errors at final testing

7. Glossary

This glossary is every major word used for this project.

Agile Methodology – A project management approach that uses short development cycles (sprints) to allow flexibility, continuous improvement, and iterative development.

Alert System – A feature that notifies students or advisors when academic performance falls below defined thresholds (e.g., low GPA or insufficient credits).

Backend – The server-side portion of the application responsible for processing data, calculations (such as GPA), and system logic.

Credit Hours – Units that represent the amount of academic coursework completed toward a degree.

Dashboard – The main user interface screen displaying academic progress, GPA, credit totals, and alerts.

Database – A structured collection of stored data (students, courses, grades, degree requirements) used by the system.

Degree Audit – An automated process that compares completed coursework against degree requirements to determine progress toward graduation.

Frontend – The visual and interactive portion of the web application that users see and interact with.

GPA (Grade Point Average) – A numerical representation of a student's academic performance calculated from course grades and credit hours.

Milestone – A significant checkpoint in the project timeline marking the completion of a major task.

Scope – The defined boundaries of what the project will and will not include.

Sprint – A short, time-boxed development cycle used in Agile methodology to complete specific tasks.

System Integration – The process of combining frontend, backend, and database components, so they function together as one system.

User Testing – The process of having end users test the system to ensure usability and functionality meet requirements

8. References

This part is a list of all the references that were needed to complete this project outline

Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® Guide)* (7th ed.). Project Management Institute.

Beck, Kent, et al. (2001). *Manifesto for Agile Software Development*. Agile Alliance. https://agilemanifesto.org

Agile Process slides

understand what the software will do.

Server – a computer that provides services to another computer.

Specification Document – A detailed statement outlining the software's

functionality and its constraints.

SQL – Structured Query Language, a standard language used to manipulate databases.

Team leader – Leader of the software development group. **Time**

line – schedule detailing a chronological sequence of events.